Student Name	ANSWERS				
Teacher	AHU	COZ	GSL	HAL	MDI

CHEMISTRY
Unit 4, SAC 1
Covering concepts from Chapters 10, 11 & 12 of Heinemann Chemistry 2
Organic Chemistry

10 September 2021, Period 3

Writing time: 45 minutes

Structure of SAC

Number of questions	Number of questions to be answered	Number of marks
6	6	26

- Students are permitted to bring into the assessment session:
  - Laptop computers, Data Book (electronic or printed)

Materials supplied
- Question and answer book of 9 pages by assignments on Teams.
- Data Book

Instructions
- Write your **name** and highlight your **teacher code** in the space provided above on this page.
- All written responses must be in English.
- Answer all questions in the spaces provided.

At the end of the assessment
- Ensure you have saved your work in Word (Ctrl + s) and then "Turned in" your task on Microsoft Teams.

Students agree that all work completed in this SAC is their own work and they have not been assisted in the task by another individual.

Students must not disclose the contents of the task; to do so will be a breach of VCE guidelines and will be dealt with according to VCAA regulations.

Instructions

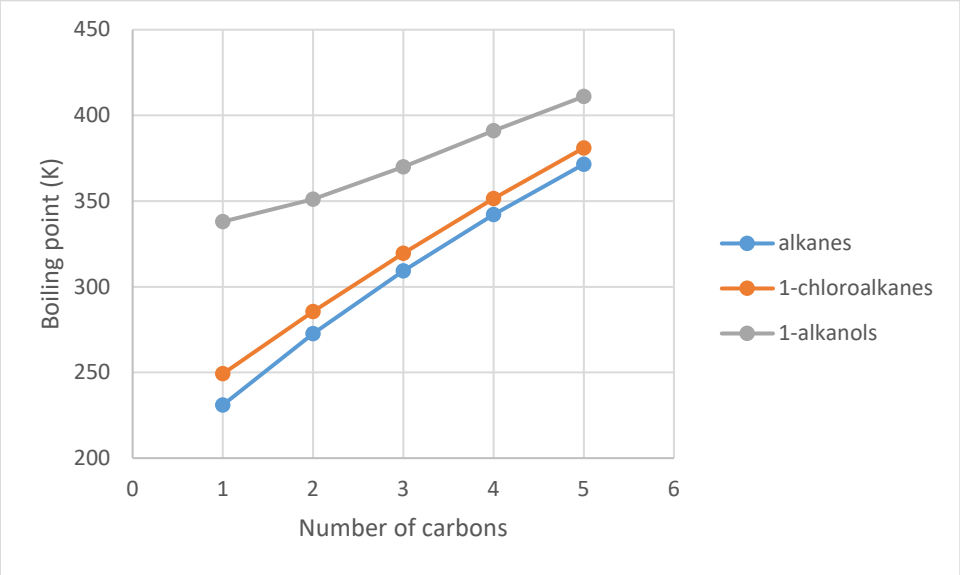
Answer **all** questions in the spaces provided.
To obtain full marks for your responses you should:

- Give simplified answers with an appropriate number of significant figures to all numerical questions; unsimplified answers will not be given full marks.
- Show all working in your answers to numerical questions. No credit will be given for an incorrect answer unless it is accompanied by details of the working.
- make sure chemical equations are balanced and that the formulas for individual substances include an indication of state; for example, $H_2(g)$; NaCl(s)

Question 1 (4 marks)

The following graph shows the boiling points of the first 5 alkanes, 1-chloroalkanes and 1-alkanols.

Graph 1: Boiling points of alkanes, 1-chloroalkanes and 1-alcohols



a. Explain why pentane has a lower boiling point than 1-chloropentane, with reference to structure and bonding. 2 mars

Answer:

Pentane molecules interact with **dispersion forces only** which require **less energy to disrupt** than the **dipole-dipole** + dispersion forces in 1-chloropentane.

b. Explain, with reference to structure and bonding, why the difference in boiling point between 1-alcohols and alkanes decreases as the number of carbons increases. 2 marks

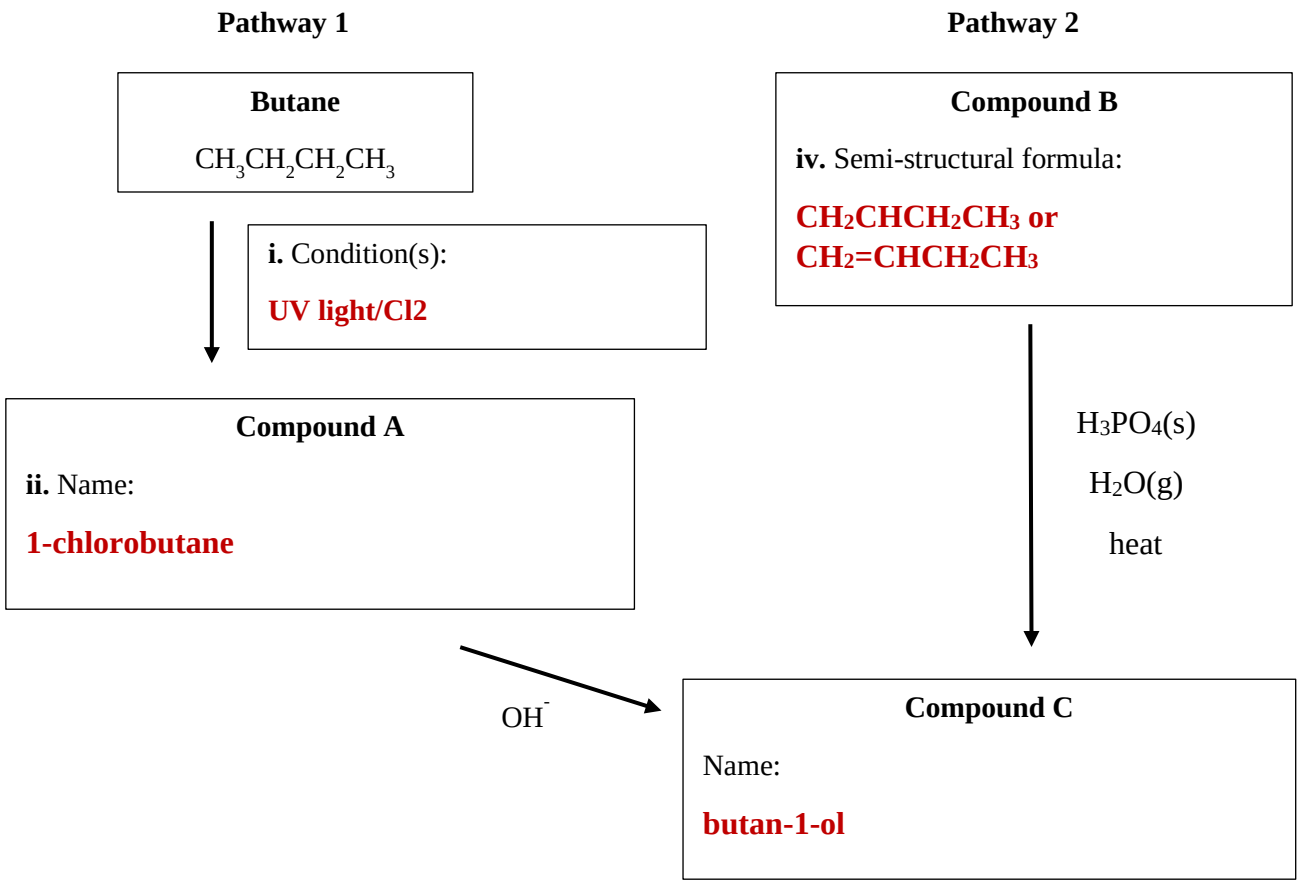
Answer:

As the number of carbons, or molar mass increases, the non-polar region of the molecule and **strength of dispersion forces increases**. (1m)

The **effect of hydrogen bonding** due to the hydroxyl **in the 1-alcohol becomes less significant** and therefore the difference in boiling point between alkanes and 1-alcohols decreases (1m).

Question 2 (6 marks)

The following diagrams represent two different pathways for the synthesis of a primary alcohol, **Compound C**.



a. Complete the pathways above by doing the following:

i. **Pathway 1** In the box provided, identify the condition(s) required for this reaction. 1 mark

ii. Name **Compound A** in the box provided. 1 mark

iii. **Compound A** can then react with a strong base, OH^-, to form the desired final product, **Compound C**. Write the name of **Compound C** in the box provided. 1 mark

iv. **Pathway 2 Compound B** is treated with water in the presence of phosphoric acid and heat to form **Compound C**. Identify **Compound B** by providing its semi-structural formula in the box provided. 1 mark

b. Propan-1-amine, a primary amine, can be synthesised using two possible reaction pathways.

Pathway 1: $CH_3CH_2CH_2OH + NH_3 \xrightarrow{\text{alumina}} CH_3CH_2CH_2NH_2 + H_2O$

Pathway 2: $CH_3CH_2CH_2Cl + NH_3 \longrightarrow CH_3CH_2CH_2NH_2 + HCl$

The atom economy for pathway 1 is 77%. Compare Pathway 1 and Pathway 2 in terms of their atom economy. 2 marks

Answer:

$M(CH_3CH_2CH_2NH_2)_{\text{desired product}}$ = 59 g/mol

$M(CH_3CH_2CH_2Cl + NH_3)_{\text{pathway 2 reactants}}$ = 95.5 g/mol

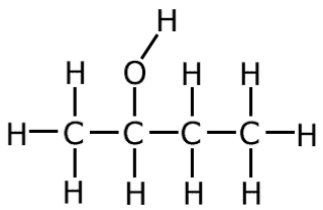
Pathway 2 atom economy = (59/95.5)*100 = **62% (1m) (not sig fig question)**

The atom economy for pathway 2 (62%) is **lower than** the atom economy for pathway 1 (77%). **(1m)**

A greater mass of reactant atoms end up in the desired product (propan-1-amine) in pathway 1.

Question 3 (3 marks)

a. The following alcohol has the formula $C_4H_{10}O$.



Explain if there could be enantiomers (optical isomers) for this alcohol. 2 marks

Answer:

C_2 (or the C bound to –OH) is attached to 4 different groups and is therefore **a chiral centre (1m)** *(Chiral centre must be identified)***.** It is the only chiral centre.

As this molecule contains one chiral centre, there could be another molecule wilt exactly the same molecular and semi-structural formulas but with a different 3D arrangement of groups around the chiral centre. There **could be enantiomers (optical isomers)** for this molecule. **(1m)**

b. Consider the list of alkenes in the following table. Indicate which alkene(s) can exist as a cis-isomer by inserting an "X" in the appropriate box. 1 mark

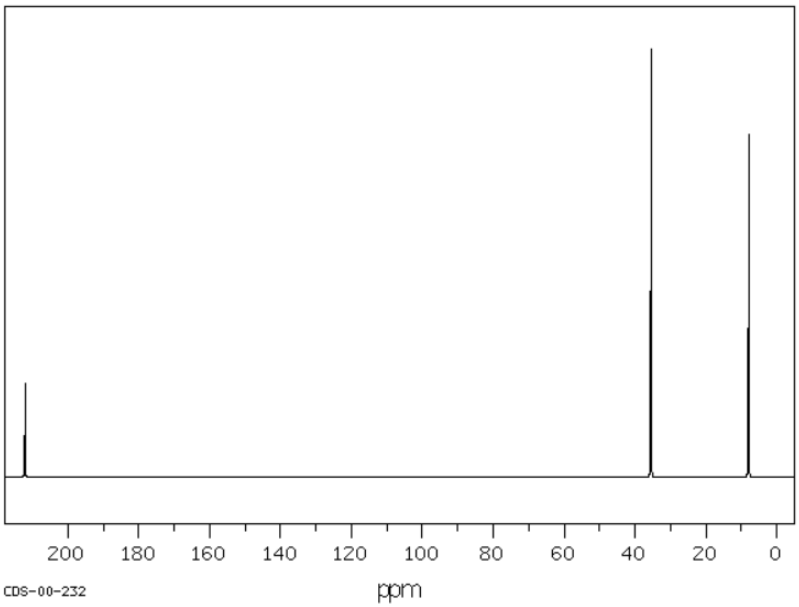
hex-2-ene	
2-methylhex-2-ene	

3-methylhex-2-ene	**X**

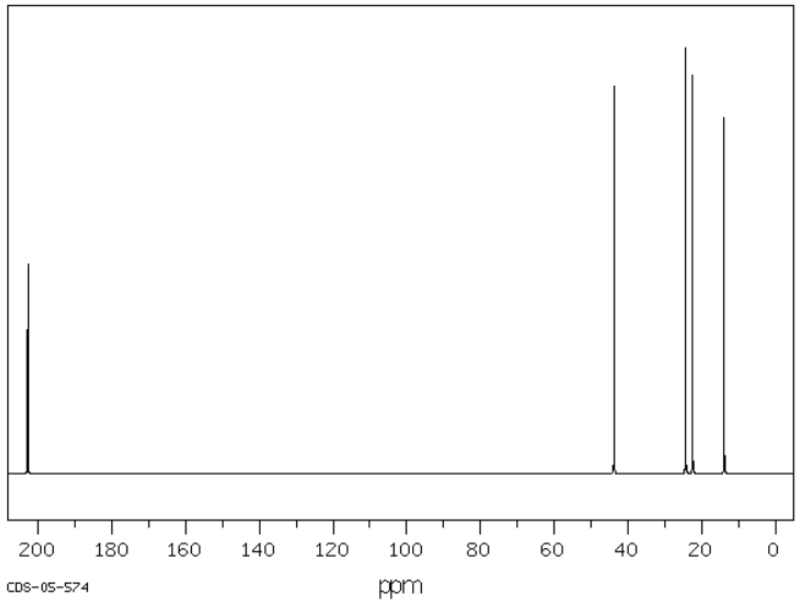
Question 4 (4 marks)

Below are the ^{13}C-NMR spectra for two different **linear** chemicals with the molecular formula $C_5H_{10}O$. **One of the molecules is an aldehyde and other one is a ketone.**

^{13}C NMR spectrum A



^{13}C NMR spectrum B



a. i. Identify whether an aldehyde or a ketone produced: 1 mark

	Aldehyde or ketone?
^{13}C NMR spectra A	**Ketone**

^{13}C NMR spectra B	**Aldehyde**

ii. Justify your responses to question 5a (above) using information from **^{13}C NMR spectra A** and **^{13}C NMR spectra B.** 1 mark

Answer:

The carbonyl in an aldehyde must be on carbon number 1. In this position and **with 5 carbons in the molecular formula, the minimum number of carbon environments for a linear aldehyde is 5 (Spectrum B). (1m)**

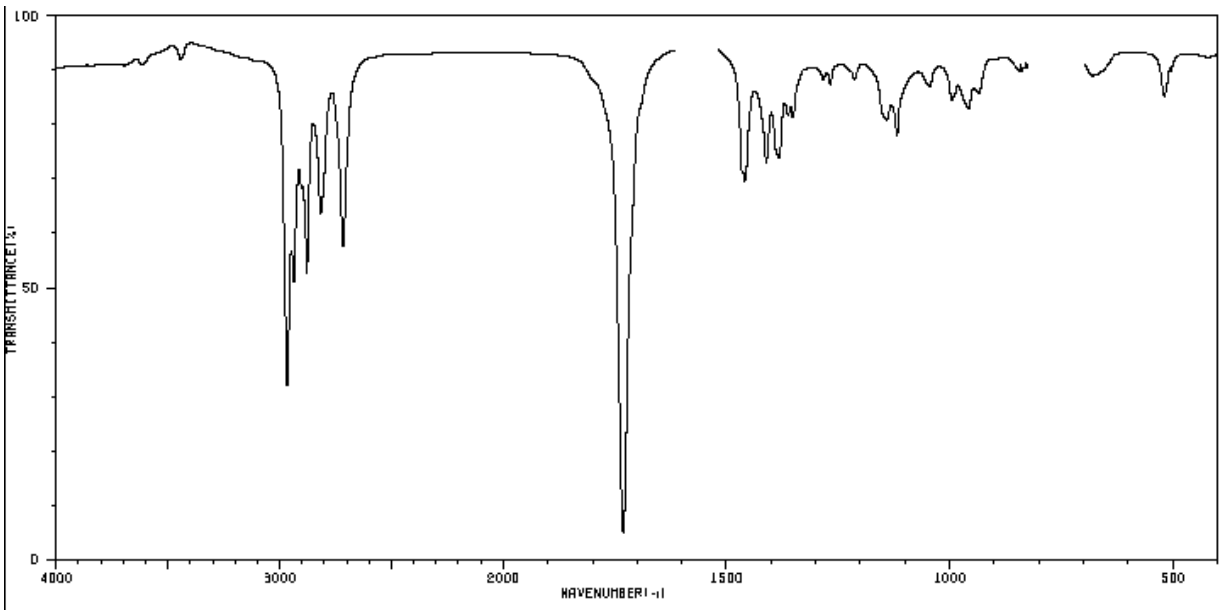
OR

With 5 carbons in the molecular formula, **the only way there can be 3 C environments (spectrum A)in a linear molecule is with symmetry** where the carbonyl carbon is in the centre – a ketone. **(1m)**

OR

The aldehyde would have an expected chemical shift of **190-200** consistent with the Spectrum B peak at ~200ppm and the ketone would have an expected chemical shift of **205-220** which is consistent with the Spectrum A peak at ~215ppm. (**Must provide supporting data from Data book**)

b. The IR spectrum for the **aldehyde** with molecular formula C_4H_8O is shown below. Identify two key absorption bands by completing the table below. 2 marks

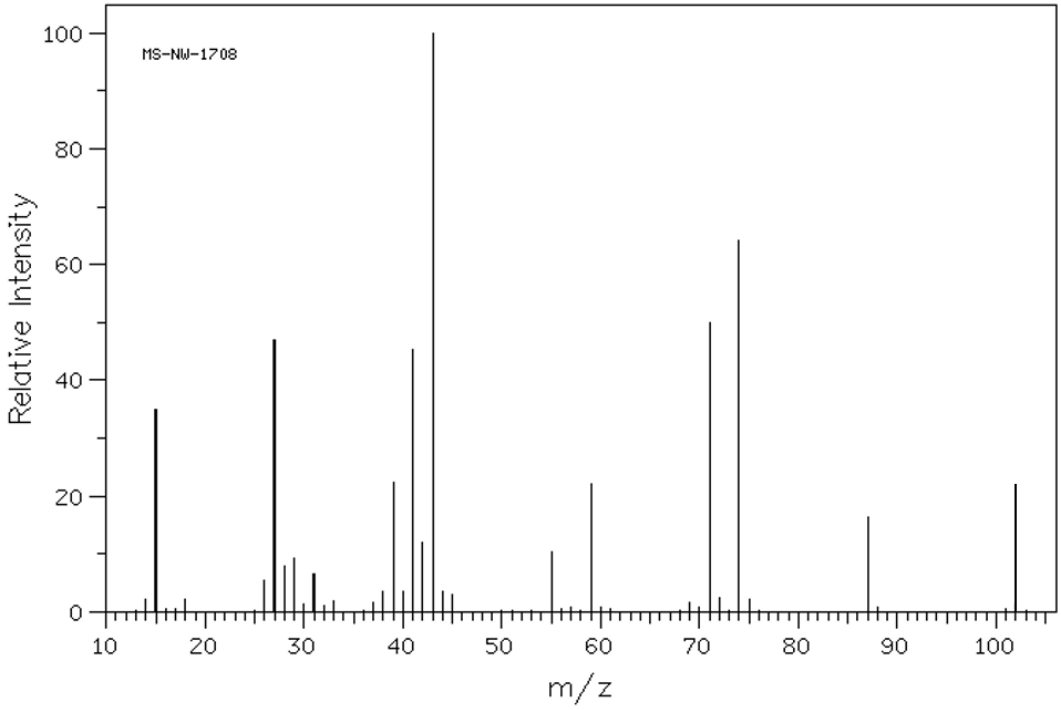


Absorption band	Bond	Wave number (cm^{-1})
1	**C=O** (aldehyde)	**~1730 (DB range 1660-1745)**
2	**C-H**	**~3000 (DB range 2850-3090)**

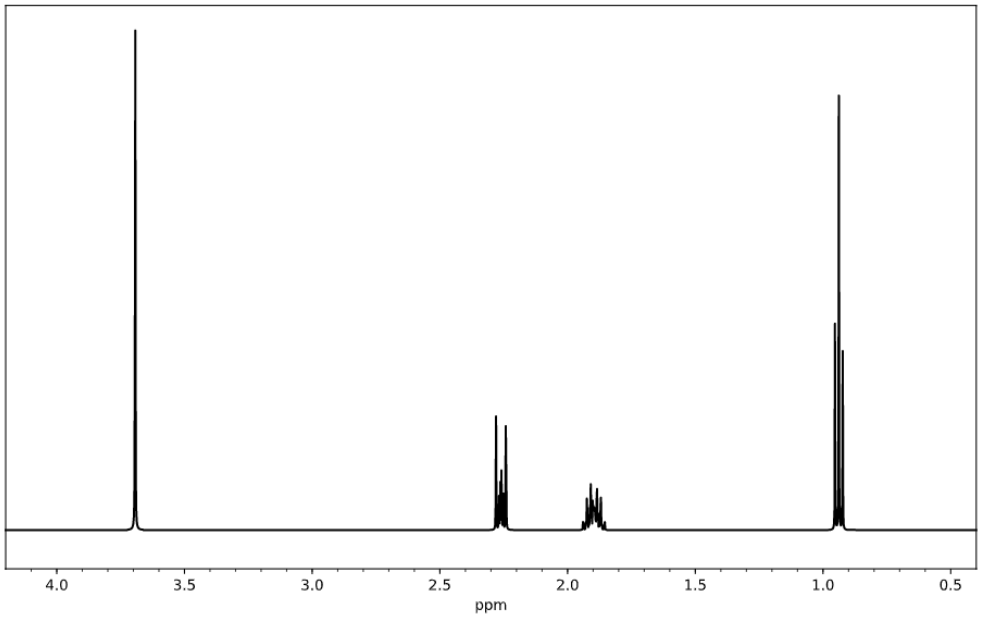
Question 5 (9 marks)

The Mass Spectrum and ^{1}H NMR Spectrum of an unknown compound with molecular formula $C_5H_{10}O_2$ is shown below.

Mass Spectrum



^{1}H NMR Spectrum



Chemical shift	Peak splitting	Relative peak area
0.9	triplet	3
1.9	sextet	2
2.3	triplet	2
3.7	singlet	3

a. The **IR Spectrum** for the unknown compound shows a strong narrow absorption band at around 1760cm^{-1}. There are no other significant bands outside of the fingerprint region.

i. Identify the bond(s) likely to be responsible for this band. 1 mark

Answer:

C=O (ester) (1m)

There are 2 oxygens in the molecular formula so ester or carboxylic acid. Wave number is too high for a carboxylic acid.

ii. Refer to the physical principles of this technique in order to explain how this band was formed. 2 marks

Answer:

Covalent bonds can vibrate and these **vibrational modes can be excited with infrared radiation.(1m)** Different bonds with different bond energies, and therefore **different functional groups, will absorb specific frequencies of IR radiation. Each specific frequency of IR absorbed** will produce a band on the spectrum. **(1m)**

Must link different frequency of IR with different types of bonds in functional group for second mark.

b. **i.** How many different hydrogen environments are there? Justify your answer using information provided in the ^{1}H NMR Spectrum. 1 mark

Answer:

4 different H environments due to **4 peaks**.

ii. Identify and describe the types of hydrogen environments that produced the signals at 1.9 ppm and 3.7ppm in the ^{1}H NMR Spectrum. 2 marks

1.9 ppm: This hydrogen environment (**containing 2 H** – likely a CH_2) produced a peak that split into a sextet indicating there are **5 equivalent (neighbouring) hydrogens**. **(1m)**

3.7 ppm: This hydrogen environment (**containing 3 H** – likely a methyl group) produced a peak that split into a singlet indicating there are **0 equivalent (neighbouring) hydrogens**. **(1m)**

c. Refer to the Mass Spectrum of the unknown compound. At what mas to charge ratio (m/z) does the molecular ion appear? 1 mark

Answer:

102

d. Name the unknown chemical. ~~2~~1 marks

Answer:

methyl butanoate

END OF QUESTION AND ANSWER BOOKLET